

GNSS AI Translation Engine

Executive Summary & Technical Overview

Precision-GNSS.com — English → Spanish Automated Technical
Translation

Status: FASE 4 & 5 complete · FASE 0-3, 6-10 pending staging access

Prepared 2026-07-23
Confidential — internal project document
Repository: AI-WP-translation

1. Executive Summary

The GNSS AI Translation Engine is a purpose-built software system that automatically translates the technical content of Precision-GNSS.com from English into European Spanish, with an architecture prepared to extend to French, German, Italian and Portuguese. Unlike a generic machine-translation plug-in, it is designed specifically for technical GNSS/RTK/surveying content, where a mistranslated number, unit, or qualifier can change the factual meaning of a claim.

The system does not translate blindly. Every piece of content passes through three independent, specialised AI calls — a **Translator**, a **Technical Reviewer**, and a **Terminology Validator** — plus a deterministic local check against a curated domain glossary, before a translation is considered safe for human review. This report summarises what has been built, how it works, what technology it uses, and what it costs to run against real DeepSeek pricing, validated with real API calls and real content scraped from the live site.

Current state: the core translation engine (extraction-agnostic: Translator, Reviewer, Terminology Validator, Glossary Engine, long-content chunking) is fully implemented, covered by 60 automated tests, and has been validated against the real DeepSeek API on real precision-gnss.com content. The WordPress/WPML integration layer is designed and documented but not yet implemented, pending access to a staging copy of the site.

2. What the Tool Does

The engine sits between WordPress (running WPML and Elementor) and the DeepSeek AI API, acting as an independent, auditable layer — never a simple script that dumps text into an AI model and republishes whatever comes back. Its responsibilities are:

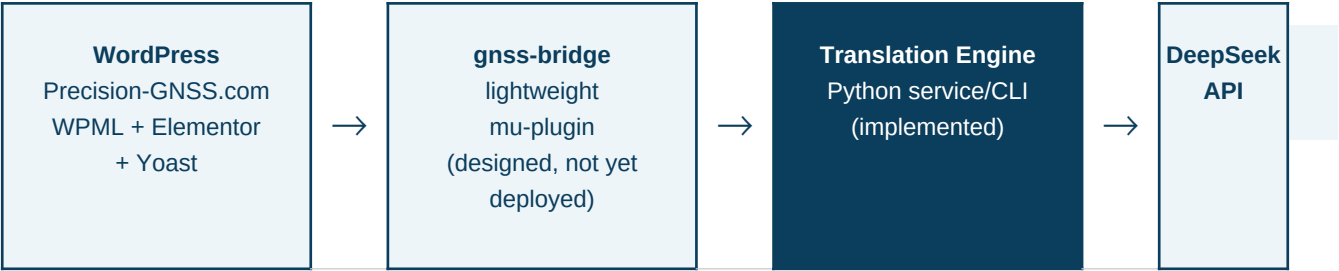
- **Extract** translatable content from WordPress pages while explicitly protecting URLs, email addresses, SKUs, model numbers, shortcodes, and API keys from being altered.
- **Translate** that content with strict technical-fidelity rules — preserving exact meaning, numbers, units, conditions, warnings, and degree of certainty.
- **Review** every translation independently for semantic drift: information added, removed, or altered compared to the source.
- **Validate terminology** against a mandatory domain glossary (GNSS, RTK, PPK, surveying, geodesy terms), both via a second AI call and a free, deterministic local check.
- **Score and gate** every translation before it is ever published — translations are created as drafts pending human approval, never auto-published.
- **Track and reuse** translations via a translation memory, so only content that actually changed gets re-translated.

3. Technology Stack

Layer	Technology	Purpose
Translation engine	Python 3.10	Core service/CLI, independent of WordPress
AI provider	DeepSeek API (deepseek-v4-pro / deepseek-v4-flash)	OpenAI-compatible chat completions, JSON mode for structured output
AI client SDK	openai Python SDK	Talks to DeepSeek via a custom <code>base_url</code>
Data validation	Pydantic v2	Strict schemas for every AI response (translation, review, terminology audit)
Testing	pytest, unittest.mock	60 automated tests, no live API calls in CI
Configuration	python-dotenv	Environment-based config, API keys never committed to source control
CMS integration (designed)	WordPress REST API + Application Passwords	Standard content read/write, authenticated per request
Multilingual layer (designed)	WPML (Multilingual CMS plan) + a custom lightweight mu-plugin (gnss-bridge)	WPML's translation-linking hooks are PHP-only; the mu-plugin exposes them safely over REST
Persistence (designed)	SQLite	Translation memory: source hash, translation, model, prompt version, QA score

4. Architecture

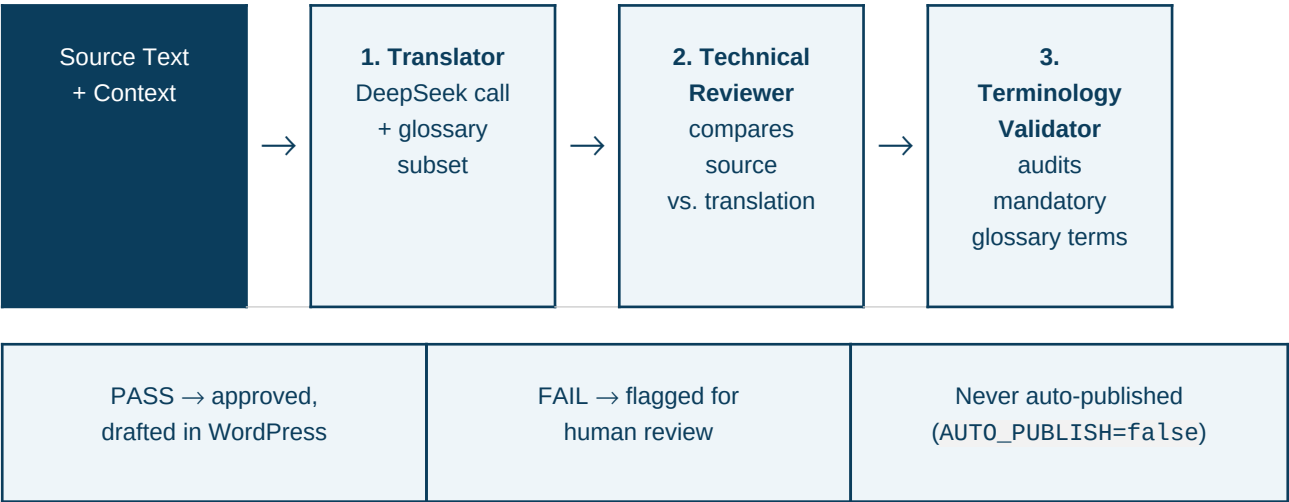
The system deliberately keeps three concerns apart: WordPress never talks to DeepSeek directly, and DeepSeek never has any awareness of WordPress. A thin, well-defined engine sits in between and owns all the intelligence — extraction rules, glossary logic, quality gates, and audit logging.



The `gnss-bridge` component exists because WPML does not expose a public REST API for creating or linking translations — that operation is only supported through internal PHP hooks (`wpml_set_element_language_details` and related filters) that must run inside WordPress itself. A minimal, auditable bridge plugin is the safest way to reach those hooks from an external Python service, without ever writing directly to WPML's database tables — which WPML's own documentation explicitly advises against.

5. Core Logic: the Translation Pipeline

Translation and quality assurance are deliberately split into independent AI calls rather than one combined request. This is not a stylistic choice — it directly reflects a validated real-world finding: during testing, the Translator alone produced a subtle but real technical error (translating *"achieves a fix within 5 seconds"* — i.e. in at most 5 seconds — as *"en menos de 5 segundos"*, i.e. in strictly less than 5 seconds). The independent Reviewer call caught this exact error and correctly failed the translation. A single combined call would very likely have missed it.



5.1 Translator

A single DeepSeek call, constrained by a strict system prompt with 15 mandatory rules (preserve exact meaning, numbers, units, conditions, warnings, certainty; never simplify, invent, or drop technical information) and 6 explicit prohibitions (never soften or strengthen claims, never translate protected product names or acronyms without instruction). The model is forced into JSON mode and must return a structured object: translation text, a 0-1 confidence score, a list of flagged issues, and the glossary terms it actually used.

5.2 Technical Reviewer

A second, independent DeepSeek call that receives both the source and the translation and checks specifically for information added, removed, or altered; changed conditions, certainty, or terminology. Returns a strict pass/fail plus a list of issues — this is the call that caught the "within 5 seconds" error described above.

5.3 Terminology Validator

A third, independent DeepSeek call auditing the translation against the mandatory glossary terms relevant to that content, flagging any deviation from the required target term.

5.4 Glossary Engine

Domain terminology (GNSS, RTK, PPK, surveying, geodesy) is stored in JSON files and only the subset of terms actually present in the current text is sent to DeepSeek on each call — never the whole glossary — keeping prompts small and costs down. A free, local, deterministic check (`validate_translation()`) also flags obviously missing mandatory terms without spending any API budget.

5.5 Chunking

Long content (a full article, a long product page) is split at paragraph boundaries — falling back to sentence boundaries only for a single oversized paragraph — so numbers, units, and technical terms are never cut mid-word. Each chunk is translated with the same shared context and glossary, then rejoined.

6. Implemented Functionality

Phase	Scope	Status
FASE 0	WordPress/WPML environment audit	Blocked — needs staging access
FASE 1	gnss-bridge mu-plugin (WPML hooks bridge)	Designed, not deployed
FASE 2	WordPress connector (Python)	Blocked — needs staging access
FASE 3	Content extraction (Elementor/Yoast)	Blocked — needs staging access

FASE 4	Translation Engine: Translator, Reviewer, Terminology Validator, chunking	Complete — 60 tests, validated with real API
FASE 5	Glossary Engine	Complete — seed glossary (10 terms), pending real-content expansion
FASE 6	Translation Memory (SQLite) & change detection	Not started
FASE 7	QA Engine: numeric/unit/URL checks, scoring system	Not started
FASE 8	WPML orchestration (end-to-end pipeline)	Blocked — needs staging access
FASE 9	Pilot on 5 real pages	Blocked — needs staging access
FASE 10	Scale-out & scheduled sync	Not started

Full detail, task-level breakdown and rationale for each phase is tracked in [ROADMAP.md](#) and [PLA-ACCIO.md](#) in the project repository, with a running session log in [LOG.md](#).

7. Cost Analysis — DeepSeek API

7.1 DeepSeek pricing (as of 2026-07-23)

Model	Input (cache miss)	Input (cache hit)	Output	Used for
deepseek-v4-pro	\$0.435 / 1M tokens	\$0.003625 / 1M tokens	\$0.87 / 1M tokens	Translation (default)
deepseek-v4-flash	\$0.14 / 1M tokens	\$0.0028 / 1M tokens	\$0.28 / 1M tokens	Optional cheaper QA model

Source: api-docs.deepseek.com/quick_start/pricing/. Both models are OpenAI-compatible and support JSON mode at no extra cost.

7.2 Methodology

Token counts are approximated at ~4 characters per token, a standard rule of thumb for English/Spanish BPE-style tokenizers. Rather than estimating on invented text, the actual system prompts produced by `app/translation/prompt_builder.py` were measured directly:

System prompt	Measured size	Approx. tokens
Translator (with 5-term glossary)	1,832 characters	~458 tokens
Technical Reviewer	832 characters	~208 tokens
Terminology Validator (with glossary)	859 characters	~215 tokens

7.3 Empirical test — real precision-gnss.com pages

Four real, public pages were scraped from the live site and run through the actual `chunk_text()` splitter and the measured prompt sizes above to produce a real, code-grounded cost estimate — not a theoretical one.

Page	Words	Characters	Chunks	Translate only (v4-pro)	Full pipeline (v4-pro)	Full pipeline (pro + flash QA)
Homepage	1,800	12,141	4	\$0.00523	\$0.01049	\$0.00692
Archaeology (RTK application)	2,156	13,871	4	\$0.00585	\$0.01173	\$0.00774
Precision Agriculture (RTK application)	3,951	24,706	7	\$0.01040	\$0.02083	\$0.01376
News: RTK/GNSS for Robotics	1,575	10,012	3	\$0.00425	\$0.00852	\$0.00562
Average	2,371	15,183	4.5	\$0.00643	\$0.01289	\$0.00851

"Full pipeline" includes all three calls (Translator + Reviewer + Terminology Validator). Cost per 1,000 characters of source content was remarkably consistent across all four pages: **\$0.00084–\$0.00086** using the full quality-assured pipeline on `deepseek-v4-pro`. Sample: `scripts/estimate_page_cost.py` in the repository reproduces these figures against any scraped page.

Practical reading: translating a typical Precision-GNSS.com content page (~2,400 words) costs roughly **\$0.006** for translation alone, or **\$0.013** with full independent quality review — well under one cent per page either way. Using the cheaper `deepseek-v4-flash` model for the two QA calls (keeping `deepseek-v4-pro` for translation itself) cuts the full-pipeline cost by about a third, to roughly **\$0.0085** per page, with no loss of translation quality since only the review layer changes model.

7.4 Site-wide projection (illustrative)

The homepage alone links to at least 13 distinct RTK application pages and several news/quick-start articles — the real total page count for the site has not yet been established (that requires the FASE 0 audit, currently blocked on staging access). For reference, at illustrative page counts using the measured full-pipeline average of \$0.0129/page:

Illustrative page count	Translate only	Full QA pipeline (pro)	Full QA pipeline (pro+flash)
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25 pages	\$0.16	\$0.32	\$0.21
50 pages	\$0.32	\$0.64	\$0.43
100 pages	\$0.64	\$1.29	\$0.85

These totals are for a full initial translation pass. Ongoing costs after launch will be far lower, since the translation memory (FASE 6) only re-translates content blocks that actually changed.

8. Current Status & Next Steps

- **Done:** Translation Engine core (FASE 4) and Glossary Engine (FASE 5) — implemented, tested (60 automated tests), and validated against the live DeepSeek API and real site content.
- **Blocked:** WordPress/WPML integration (FASES 0-3, 8-9) — requires access to a staging copy of Precision-GNSS.com, which has been requested and is pending.
- **Available now:** the `gnss-bridge` mu-plugin skeleton (FASE 1) can be written and reviewed without deployment; QA Engine (FASE 7: numeric, unit, URL, and terminology checks) and Translation Memory (FASE 6) can also proceed independently of staging access.